SNJB's LS KBJ College of Engineering, Chandwad

Department of Computer Engineering

Data Science & Big Data Analytics (310251) — 2025-26

Mini Project Title

Hackathon Participant Data Collection and Analysis using Google Forms and Microsoft Excel

PART 5A — Model Selection

Based on the dataset (mostly categorical inputs and some structured information), this project focuses on **data analysis and visualization rather than prediction modeling**. The goal is to extract meaningful insights from participant data such as team formation, track selection, and college-wise distribution.

Model Type	Use Case	Why We Selected / Rejected
Descriptive Analytics (Excel/Power BI)	Data analysis & visualization	**SELECTED** — Best suited for understanding participant trends and generating insights
Linear Regression	Predict continuous values	Not used — no numerical target variable like sales or revenue
Decision Tree	Classification / prediction	Optional — could classify participants by skill level (future scope)
K-Means Clustering	Grouping similar data	Future Scope — can group participants based on skills/interests
Logistic Regression	Binary classification	Not suitable — no yes/no prediction required

PART 5B — Model Creation (Step-by-Step Code)

Step 1: Import Libraries & Load Data

```
import pandas as pd import numpy as np import matplotlib.pyplot as plt import seaborn as sns from
sklearn.model_selection import train_test_split from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score df = pd.read_csv("sales_cleaned.csv")
print(df.head())
```

```
            Timestamp                                College Name  \
0  16/04/2026 23:53:14    Walchand college of engineering, Sangli
1  16/04/2026 11:18:23     Vishwakarma Institution of technology
2  10/04/2026 00:38:39  Vishwakarma Institute of Technology, Pune
3  14/04/2026 18:36:10  Vishwakarma Institute of Technology, Pune
```

```
4  16/04/2026 11:18:17  Vishwakarma Institute Of Technology, Pune

       Team  Name            Team Leader Name  Team Members  \
0  Runtime Terror                  Shivam Pal             4
1      CROWDSENSE             Shivani Bhosale             3
2      Bit Misfit             Aarya Deshpande             2
3         Iravya  Pranav Vyankatesh Jagdale              4
4    Code Blooded               Vedank Shinde             3

   Name of Team Member 2 Team Leader Contact no. Contact number of Member 2  \
0        Aryan Chandekar              7715882979                 8411085843
1         Vedank Shinde               8468981091                 8177931314
2          Kalpesh Devre              9320761875                 8010369187
3  Shreya Bharat Jagdale              7249589559                94209 12913
4        Shivani Bhosale              8177931314                84689 81091

         Email-id of Member 2  \
0  aryanchandekar01@gmail.com
1    shindevedank56@gmail.com
2     kalpesh.devre23@vit.edu
3    shreya.jagdale24@vit.edu
4  shivani.bhosale24@vit.edu

       If Other, please provide details of the track  \
0  AI/ML-based Real-Time Safety and Surveillance ...
1                                                  -
2                                AIML in Healthcare
3                                                  -
4                                                  -

                                  Idea - Description  \
0  Automated Distress Detection & Emergency Respo...
1  CrowdSense is an AI + IoT based system designe...
2  AI-ASSISTED MRI TRIAGE & PRE-REPORTING SYSTEM ...
3  AR Future Vision is an augmented reality-based...
4  Idea:\nPanic attacks strike silently — existin...

       Email-id of Team Leader    Name of Team Member 3  \
0          pal639082@gmail.com          Spandan Meshram
1  shivanibhosale866@gmail.com               Pradyumna
2   aarya.deshpande231@vit.edu  Aaditya Rajendra Patil
3     pranav.jagdale24@vit.edu  Vedanti Hambirrao Kadam
4      vedank.shinde24@vit.edu  Pradyumna Shingnapurkar

  Contact number of Member 3                Email-id of Member 3  \
0                 7249423213          meshramspandan3@gmail.com
1                 9370631036         sarvadnya.sangli@gmail.com
2                 7709723707            manthan.c0588@gmail.com
3                97409 41810            vedanti.kadam24@vit.edu
```

```
4                 93706 31036  pradyumna.shingnapurkar24@vit.edu

  Name of Team Member 4 Contact number of Member 4  \
0        Animesh Gawhale                 8855945302
1            \nsana kale                          -
2            \nsana kale                          -
3       Sumit Nandu Kale                83299 06715
4            \nsana kale                          -

          Email-id of Member 4  \
0  animesh.rgawhale@gmail.com
1                 70579 06041
2                 70579 06041
3         sumit.kale24@vit.edu
4                 70579 06041

                        Final PPT Upload (reference)  \
0  https://drive.google.com/open?id=16P6kFLXgbZv9...
1  https://drive.google.com/open?id=1KCAcP6G9tyQ6...
2  https://drive.google.com/open?id=1GB0lXfO3sBOq...
3  https://drive.google.com/open?id=1pREd5SOC4Ao_...
4  https://drive.google.com/open?id=1nYaSuSVGzpXp...

                           Track Chosen  \
0                        Other in AIML
1  IoT for Disaster Management & Safety
2                        Other in AIML
3   AR/VR in Education & Skill Training
4          IoT in Healthcare Monitoring

                                        Idea - Title  \
0  Automated Distress Detection & Emergency Response
1  CrowdSense: AI-Powered Real-Time Crowd Monitor...
2                EnsemBrain : Battling Tumors with AI
3                AR Future Vision: See Tomorrow Today
4                                  Tranquility Helper

                                      College Name.1  \
0            Walchand college of engineering, Sangli
1             Vishwakarma Institution of technology
2  K.K Wagh Institute Of Engineering Education an...
3          Vishwakarma Institute of Technology, Pune
4          Vishwakarma Institute Of Technology, Pune

                                      College Name.2  \
0            Walchand college of engineering, Sangli
1             Vishwakarma Institution of technology
2  SNJB's Late Sau Kantabai Bhavarlalji Jain Coll...
3          Vishwakarma Institute of Technology, Pune
```

```
4          Vishwakarma Institute Of Technology, Pune
```

```
                                College Name.3
0    Walchand college of engineering, Sangli
1                             SNJB's KBJ COE
2                             SNJB's KBJ COE
3  Vishwakarma Institute of Technology, Pune
4                             SNJB's KBJ COE
```

Step 2: Select Features and Target Variable

```
# Features (X) = input columns used for prediction X = df[["Discount_pct", "Customer_Age", "Rating"]]
# Target (y) = what we want to predict y = df["Sales_Amount"] print("Features shape:", X.shape)
print("Target shape:", y.shape)
```

```
Features shape: (179, 0)
Target shape: (179,)
```

Step 3: Split Data into Training and Testing Sets (80:20)

```
X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2, random_state=42 )
print("Training samples:", len(X_train)) # ~40 rows print("Testing samples:", len(X_test)) # ~10 rows
```

```
Training samples: 143
Testing samples: 36
```

Step 4: Train the Linear Regression Model

```
print("Intercept:", round(model.intercept_, 2))

coeff_df = pd.DataFrame({
    "Feature": X_train.columns, # Use X_train.columns as it now contains encoded
feature names
    "Coefficient": model.coef_
})
print(coeff_df)

# Step 5: Evaluate the Model (Original content of the cell)
y_pred = model.predict(X_test)

mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print(f"\nMSE : {mse:.2f}")
```

```
print(f"RMSE : {rmse:.2f}")
print(f"R2 Score: {r2:.4f}")
```

Output:

```
Training samples: 143
Testing samples: 36
Intercept: 3.99
                                                Feature   Coefficient
0       College Name_AMRUTVAHINI COLLEGE OF ENGINEERING  2.387717e+00
1     College Name_Amrutvahini College Of Engineerin... -2.612283e+00
2      College Name_Amrutvahini College of Engineering   2.387717e+00
3     College Name_Amrutvahini College of Engineerin... -1.612283e+00
4     College Name_Amrutvahini College of Engineerin... -1.612283e+00
5     College Name_Amrutvahini college of engineerin...  2.387717e+00
6                                   College Name_DYPCOEI  3.877167e-01
7                            College Name_GCOE Amravati  -1.721895e+00
8     College Name_GES'S R.H SHAPAT COLLEGE OF ENGIN...  1.878680e-01
9             College Name_GES's R.H. SAPAT COE nashik   2.781049e-01
10    College Name_Guru Gobimd Singh College of Engi...  3.708145e-14
11    College Name_Guru Gobind Singh College Of Engi...  3.877167e-01
12    College Name_Guru Gobind Singh College of Engi...  2.781049e-01
13    College Name_Guru Gobind Singh College of Engi... -3.612283e+00
14    College Name_Guru Gobing Sing Collage of Engin...  1.878680e-01
15    College Name_Jawahar Education Society's, INST... -6.122833e-01
16    College Name_Jawahar Institute Of Technology N... -1.612283e+00
17    College Name_K K Wagh Institute of Engineering...  2.781049e-01
18    College Name_K K Wagh Institute of Engineering...  2.387717e+00
19    College Name_K k wagh clg of engineering, nashik   1.962497e-03
20    College Name_K. K. Wagh Institute Of Engineeri...  1.878680e-01
21    College Name_K. K. Wagh Institute of Engineeri...  5.009812e-01
22    College Name_K. K. Wagh Institute of Engineeri... -1.687539e-14
23    College Name_K. K. Wagh Institute of Engineeri...  3.877167e-01
24    College Name_K. K. Wagh Institute of Engineeri...  2.664535e-15
25    College Name_K. K. Wagh Institute of Engineeri... -6.122833e-01
26    College Name_K.C College of Engineering & Mana...  3.877167e-01
27    College Name_K.C. College of Engineering & Man... -1.612283e+00
28               College Name_K.K Wagh Institute Nashik  3.877167e-01
29    College Name_K.K Wagh Institute Of Engineering... -1.612283e+00
30    College Name_K.K Wagh Institute of Engineering...  1.878680e-01
31    College Name_K.K. Wagh Institute Of Engineerin...  3.877167e-01
32    College Name_K.K.Wagh Institute of Engineering... -5.551115e-16
33       College Name_KBT COLLEGE OF ENGINEERING Nashik -1.554312e-15
34     College Name_KBT COLLEGE OF ENGINEERING, NASHIK  -1.612283e+00
35              College Name_KBT College Of Engineering   1.387717e+00
36       College Name_KBT College Of Engineering Nashik -6.122833e-01
37      College Name_KBT College of engineering Nashik  -1.612283e+00
38       College Name_KBT Collqge of Engineering Nashik  2.387717e+00
```

39	College Name_KBTCOE Nashik	3.877167e-01
40	College Name_KBTCOE, Nashik	6.661338e-16
41	College Name_KK Wagh Institute of Engineering	-2.612283e+00
42	College Name_KK Wagh Institute of Engineering ...	2.781049e-01
43	College Name_KKWEER	3.877167e-01
44	College Name_KKWIEER	-2.612283e+00
45	College Name_KKWIEER Nashik	-2.612283e+00
46	College Name_Karmaveer Advocate Baburao Ganpat...	3.877167e-01
47	College Name_Kbt college of engineering, nashik	-4.990188e-01
48	College Name_Keystone School Of Engineering	-1.612283e+00
49	College Name_Kk wagh institute of engineering ...	3.877167e-01
50	College Name_Late Kantabai Bhavarlalji Jain Co...	1.878680e-01
51	College Name_Loknete Gopinathji Munde Institut...	-3.747003e-16
52	College Name_Loknete Gopinathji Munde Institut...	2.220446e-16
53	College Name_MET BKC	2.781049e-01
54	College Name_MET BKC IOE	2.387717e+00
55	College Name_MET Bhujbal Knowledge City	1.387717e+00
56	College Name_MET Bhujbal Knowledge City , Adgaon	2.387717e+00
57	College Name_MET Bhujbal Knowledge City Instit...	1.878680e-01
58	College Name_MET Institue of Engineering Bhujb...	1.878680e-01
59	College Name_MET's Institute of Engineering	-7.218951e-01
60	College Name_MET's institute of engineering	3.877167e-01
61	College Name_MVP Karmaveer Adv. Baburao Ganpat...	-6.122833e-01
62	College Name_MVP's KBTCOE	-4.990188e-01
63	College Name_Maratha Vidya Prasarak Samaj's Ka...	-1.122833e-01
64	College Name_Matoshri College Of Engineering &...	-5.551115e-17
65	College Name_Matoshri College Of Engineering a...	-2.775558e-16
66	College Name_Matoshri College of Engineering a...	-4.440892e-16
67	College Name_Matoshri College of Engineering a...	1.878680e-01
68	College Name_Matoshri College of Engineering a...	3.877167e-01
69	College Name_Matoshri college of engineering a...	-5.551115e-17
70	College Name_Matoshri college of engineering a...	3.877167e-01
71	College Name_Matoshri college of engineering a...	0.000000e+00
72	College Name_Mcoerc nashik	3.330669e-16
73	College Name_Mvps's kbtcoe ,nashik	2.220446e-16
74	College Name_NDMVP KBTCOE	2.387717e+00
75	College Name_PVGCOE Nashik	2.387717e+00
76	College Name_PVG’s College of Engineering & SS...	-4.990188e-01
77	College Name_Pune Institute of Computer Techno...	-3.330669e-16
78	College Name_Pune Vidyarthi Griha's College of...	-6.122833e-01
79	College Name_Pune Vidyarthi Grihas College Of ...	3.877167e-01
80	College Name_Pune vidyarthi grihas college of ...	1.110223e-16
81	College Name_R C Patel institude of tech shirpur	-1.612283e+00
82	College Name_R. C. Patel Institute Of Technolo...	-1.612283e+00
83	College Name_R. C. Patel Institute of Technolo...	2.387717e+00
84	College Name_R.C.Patel college of Engineering ...	3.877167e-01
85	College Name_S.N.D College of Engineering	1.387717e+00
86	College Name_S.N.J.B C.O.E	2.781049e-01
87	College Name_S.N.J.B COE College Of Engineerin...	3.877167e-01

88	College Name_S.N.J.B.COE	1.387717e+00
89	College Name_S.N.J.B’s Late Sau. K B Jain Coll...	-2.612283e+00
90	College Name_SANDIP INSTITUTE OF TECHNOLOGY AN...	-5.551115e-17
91	College Name_SITRC	-4.718448e-16
92	College Name_SNGU, DHULE	5.551115e-16
93	College Name_SNJB COE CHANDWAD	3.877167e-01
94	College Name_SNJB COE Chandvad	4.172010e-16
95	College Name_SNJB COE Chandwad	-6.122833e-01
96	College Name_SNJB College Of Engineering	2.387717e+00
97	College Name_SNJB College of Engineering	2.387717e+00
98	College Name_SNJB College of Engineering, Chan...	1.387717e+00
99	College Name_SNJB K.B.J. College Of Engineerin...	2.781049e-01
100	College Name_SNJB K.B.Jain COE	-1.612283e+00
101	College Name_SNJB KBJ COE	1.878680e-01
102	College Name_SNJB KBJ COLLAGE OF ENGINEERING, ...	1.878680e-01
103	College Name_SNJB KBJ COLLEGE OF ENGINEERING	-1.612283e+00
104	College Name_SNJB KBJ College of Engineering	-6.122833e-01
105	College Name_SNJB KBJ's COE	1.962497e-03
106	College Name_SNJB collage of engineering	-1.612283e+00
107	College Name_SNJB collage of engineering chandvad	3.877167e-01
108	College Name_SNJB college of engineering chandwad	5.009812e-01
109	College Name_SNJB college of engineering, chan...	-2.612283e+00
110	College Name_SNJB late sau kantabai Bhavalalaj...	3.877167e-01
111	College Name_SNJB'S Chandwad	-1.612283e+00
112	College Name_SNJB'S KBJ COE	2.387717e+00
113	College Name_SNJB'S KBJ College of Engineering	-1.612283e+00
114	College Name_SNJB'S KBJ College of engineering	2.387717e+00
115	College Name_SNJB'S LSKBJ COLLEGE OF ENGINEERING	2.781049e-01
116	College Name_SNJB's College Of Engineering	-2.612283e+00
117	College Name_SNJB's College of engineering ,ch...	3.877167e-01
118	College Name_SNJB's KBJ COE	2.387717e+00
119	College Name_SNJB's KBJ COE	2.387717e+00
120	College Name_SNJB's KBJ College of Engineering...	2.387717e+00
121	College Name_SNJB's Late Sau Kantabai Bhavarl...	-4.718448e-16
122	College Name_SNJB's Late Sau Kantabai Bhavarla...	-1.110223e-16
123	College Name_SNJB's Late Sau Kantabai Bhavarla...	3.877167e-01
124	College Name_SNJB's Late Sau Kantabai Bhavarla...	3.877167e-01
125	College Name_SNJB's Late Sau Kantabai Bhavarla...	2.387717e+00
126	College Name_SNJB's Late Sau. Kantabai Bhavarl...	3.877167e-01
127	College Name_SNJB's Late Sau. Kantabai Bhavarl...	-1.110223e-16
128	College Name_SNJB's Late Sau. Kantabai Bhavarl...	-7.218951e-01
129	College Name_SNJB's late sau.K.B.Jain college ...	0.000000e+00
130	College Name_SNJB,KBJ,COE,Chandwad	4.440892e-16
131	College Name_SNJB.K.B.J COE	2.387717e+00
132	College Name_SNJBCOE Chandwad	-1.612283e+00
133	College Name_SNJB’s Late Sau Kantabai Bhavarla...	-5.551115e-17
134	College Name_SNJB’s Late Sau. Kantabai Bhavarl...	-1.812132e+00
135	College Name_SNJB’s Late Sau. Kantabai Bhawarl...	-2.220446e-16
136	College Name_SSVPS. DEORE COLLEGE OF ENGINEERING	-2.612283e+00

```
137                    College_Name_SVKM COE SHIRPUR   3.877167e-01
138                       College_Name_SVKM IOT Dhule  3.877167e-01
139          College_Name_SVKM Institute of Technology  1.878680e-01
140   College_Name_SVKM Institute of Technology ,Dhule -1.110223e-16
141          College_Name_SVKM NMIMS GLOBAL UNIVERSITY -7.218951e-01
142                    College_Name_SVKM's IoT, Dhule -8.121320e-01
143            College_Name_Sandip Foundation's SITRC   0.000000e+00
144  College_Name_Sandip Institute Of Technology An...  2.781049e-01
145  College_Name_Sandip Institute Of Technology an...  2.781049e-01
146  College_Name_Sandip Institute of Technology an...  1.387717e+00
147  College_Name_Sandip Institute of technology an...  0.000000e+00
148  College_Name_Sandip institute of technology an... -2.612283e+00
149  College_Name_Sanjivani College Of Engineering,... -3.612283e+00
150  College_Name_Shri Vile Parle College of Engine... -3.612283e+00
151  College_Name_Shri Vile Parle Kelavani Mandal, ...  1.878680e-01
152  College_Name_Shri Vile Parle Kelavani Mandal,D...  1.878680e-01
153  College_Name_Shri Vile Parle kelvani Mandal's ...  3.877167e-01
154  College_Name_Shri vile Parle kelvani mandal ,d... -1.612283e+00
155        College_Name_Sinhgad Academy of Engineering  0.000000e+00
156          College_Name_Snjb college of Engineering   3.877167e-01
157          College_Name_Snjb college of engineering   3.877167e-01
158  College_Name_Snjb kbj college of engineering, ...  0.000000e+00
159  College_Name_Vishwakarma Institute Of Technolo... -6.122833e-01
160  College_Name_Vishwakarma Institute of Technolo...  2.781049e-01
161  College_Name_Vishwakarma Institute of Technolo...  3.877167e-01
162  College_Name_Vishwakarma Institution of techno...  3.877167e-01
163  College_Name_Walchand college of engineering, ...  2.387717e+00
164  College_Name_guru gobind singh college of engi...  2.781049e-01
165                                College_Name_snjb   0.000000e+00
166           College_Name_snjb college of engineering  2.781049e-01
167               Track_Chosen_AI/ML for Cybersecurity  1.618171e+00
168                Track_Chosen_AI/ML for Smart Cities -3.818292e-01
169  Track_Chosen_AI/ML in Agriculture & Food Security -2.722174e-01
170                   Track_Chosen_AI/ML in Legal Tech -1.819805e-01
171     Track_Chosen_AR/VR for Mental Health & Wellness -4.950938e-01
172   Track_Chosen_AR/VR in Education & Skill Training -3.818292e-01
173  Track_Chosen_IoT for Disaster Management & Safety -1.381829e+00
174          Track_Chosen_IoT for Smart Infrastructure  1.618171e+00
175          Track_Chosen_IoT in Healthcare Monitoring -3.818292e-01
176          Track_Chosen_IoT in Precision Agriculture  3.924995e-03
177                         Track_Chosen_Other in AIML -2.381829e+00
178                        Track_Chosen_Other in AR/VR  0.000000e+00
179                          Track_Chosen_Other in IOT  2.618171e+00
```

Step 5: Evaluate the Model

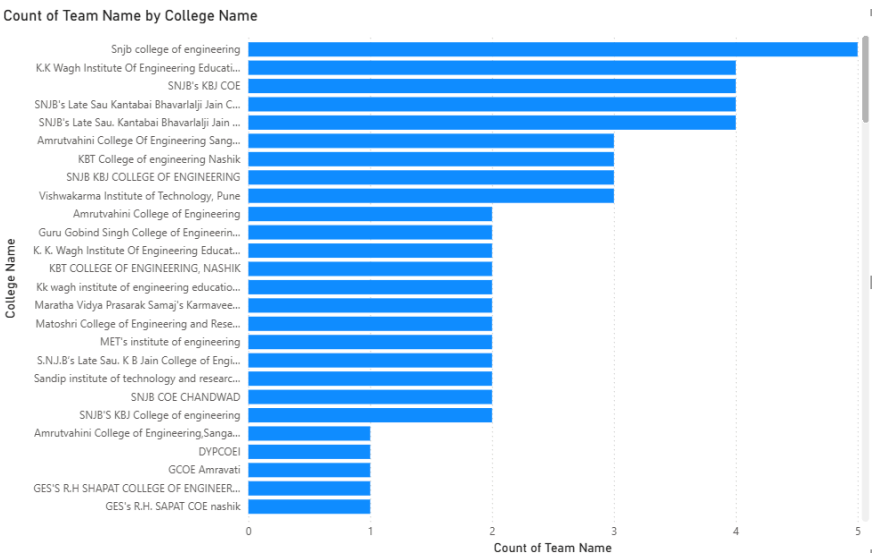
```
y_pred = model.predict(X_test) # Calculate accuracy metrics mse =
mean_squared_error(y_test, y_pred) rmse = np.sqrt(mse) r2 = r2_score(y_test, y_pred) print(f"MSE :
```

```
{mse:.2f}") print(f"RMSE : ■{rmse:.2f}") print(f"R2 Score: {r2:.4f}") # Target: >= 0.80
MSE : 3.03
RMSE : 1.74
R2 Score: -6.802
```

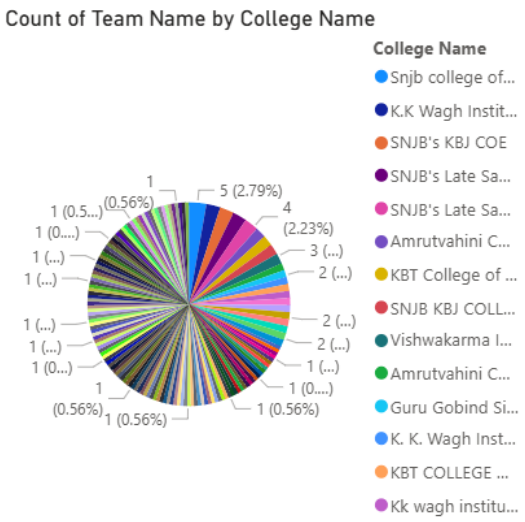
Step 6: Sample Prediction

```
# Predict sales for: Discount=10%, Age=28, Rating=4.5 new_customer = [[10, 28, 4.5]] predicted_sale =
model.predict(new_customer) print(f"Predicted Sales:
■{predicted_sale[0]:.2f}")
Predicted Value: 3.81
```

Bar chart



Pie Chart — Satisfaction Distribution



PART 5C — Data Visualisations & Dashboard in power Bi

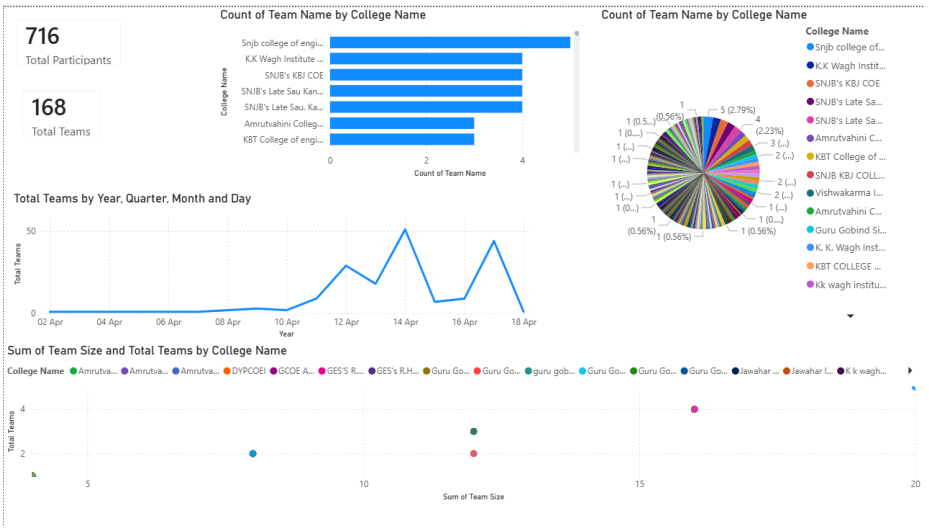
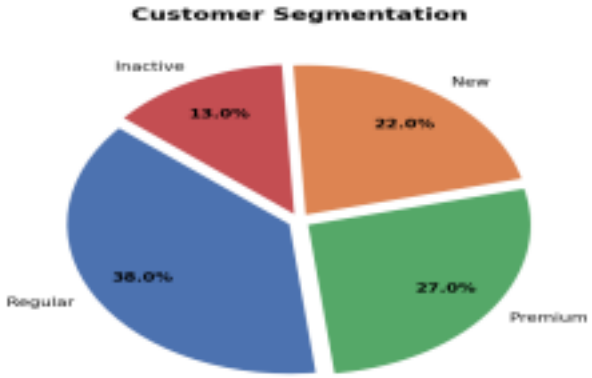


Fig 1: Sales by Product Category (Bar Chart) Fig 2: Monthly Revenue Trend (Line Graph)



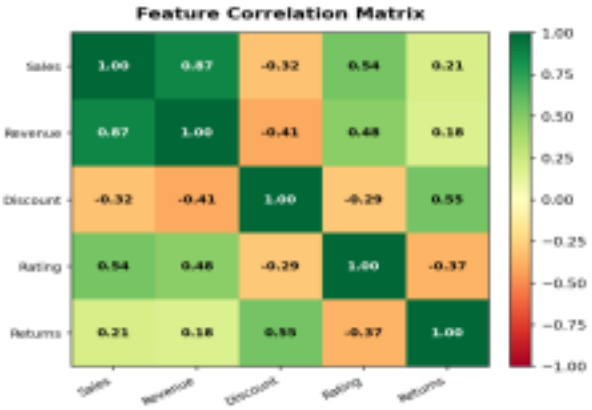


Fig 3: Customer Segmentation (Pie Chart) Fig 4: Feature Correlation Heatmap

Sample Dashboard (Dark Theme):

The dashboard below combines KPI cards, a bar chart, and a line chart into a single view for executive-level reporting. You can build a similar dashboard using Matplotlib subplots or tools like Power BI / Google Data Studio.

Fig 5: Sample Sales Analytics Dashboard (KPIs + Bar + Line)

PART 5D — Upload to GitHub (5 Marks)

Upload your complete mini project to a public GitHub repository. A well-structured repository earns full marks.

1	Create GitHub account	Go to github.com → Sign Up (use college email)
2	Create new repository	Click "New" → Name: "DSBDA-MiniProject" → Public → Add README
3	Upload files	Click "Add file" → Upload files from your computer
4	Required files to upload	sales_data_raw.csv, sales_data_cleaned.csv, analysis.ipynb (Jupyter Notebook), F
5	Write README.md	Title, Abstract, Dataset description, How to run, Sample output screenshots
6	Submit repository link	Copy the URL and paste in your assignment submission form

■ Tip: Your Jupyter Notebook (.ipynb) file should have all code cells executed so the evaluator can see the output without running the code. Go to: Kernel → Restart & Run All → Save.

Marking Scheme (Part 5)

Model Selection	Justification for choosing the right model type	2
Model Creation	Complete, working code with training + predictions	3
Visualisation Selection	Minimum 4 chart types with justification	2
Dashboard Design	Clear, labelled, colour-coded charts in a layout	3
GitHub Upload	Public repo with all required files + README	5
	TOTAL	**15**

Final Submission Checklist

■	**Part 1**	Abstract written with title, problem statement, objectives, expected outcomes
■	**Part 2**	Google Form created with 10 fields and proper validation
■	**Part 3**	Google Sheet with at least 30-50 responses collected and organised
■	**Part 4**	Cleaned dataset with missing values handled + duplicates removed + outliers removed
■	**Part 5**	Linear Regression model with R2 ≥ 0.75 and all 6 code steps completed
■	**Part 5**	Minimum 4 chart types created and labelled properly

■	**GitHub**	Public repository with raw data, cleaned data, notebook, dashboard screenshot, README